

From Raw ERP Data to Executive Decision Support

How integrated sales, inventory, and profitability analytics improve business decision-making

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A practical case study in transforming ERP-sourced data into reusable, management-ready reporting.

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Turning existing data into clearer business decisions

Organizations often have substantial operational and financial data inside ERP systems, yet leaders still struggle to answer basic questions about product performance, inventory exposure, customer-specific programs, and profitability. The issue is rarely data availability. The issue is converting raw transactions into a structure that supports timely decisions.

This paper describes an anonymized analytics engagement that began with a Power BI download sourced from an ERP platform. Raw records were cleaned, summarized, classified, and connected across inventory, sales, cost, volume, and gross margin. The result was a reusable analytical table capable of producing business-unit reports and answering specific requests without rebuilding the analysis each time.

BUSINESS NEED	ANALYTICAL RESPONSE
Leadership needed a clearer view of sales, inventory, and profitability by business unit and product category.	A structured dataset combined item, inventory, sales, cost, and gross-margin measures in one reporting layer.
Customer-specific and private-label products were not always consistently classified.	Supplemental product and customer logic created a repeatable classification approach.
Ad hoc requests required manual data gathering and repeated report creation.	The reusable model could be filtered, pivoted, and exported into decision-ready reports.

The core value was not another spreadsheet. It was a reusable analytic foundation that converted recurring questions into repeatable decision support.

Key outcomes: improved visibility, faster response to leadership, consistent metrics, integrated inventory and profitability reporting, and lower dependence on one-time manual analysis.

This document uses anonymized descriptions and synthetic chart data. No customer names, company names, item numbers, prices, costs, or confidential financial results are presented.

Data existed, but the answer did not

A business leader requested detailed information on customer-specific and private-label products. The desired output required more than a standard sales report. The business needed a single view that connected the product portfolio to current inventory, historical volume, revenue, cost, and gross margin.

The source environment contained the necessary information, but the records originated as a large Power BI download fed from the ERP system. The raw output was too detailed for management use and included thousands of item-level records, multiple organizational hierarchies, product descriptions, cost measures, customer relationships, and sales measures.

The reporting problem therefore had four parts:

- Structure the raw download into consistent item-level measures.
- Create business-unit, product-family, and customer-specific classifications.
- Connect on-hand inventory to historical sales, cost, and gross margin.
- Produce targeted reports quickly when sales or operations leaders requested them.

The required analytical fields

Product and classification	Inventory	Performance
Item group	On-hand quantity	Prior-year volume
Product description	On-hand value	Prior-year sales and cost
Regular / private-label logic	Average unit cost	Current-year volume
Business-unit hierarchy	Standard cost	Current-year sales, cost, and GM%

Traditional reporting asks, “What happened?” Integrated analytics also asks, “Where did it happen, why does it matter, and what should management examine next?”

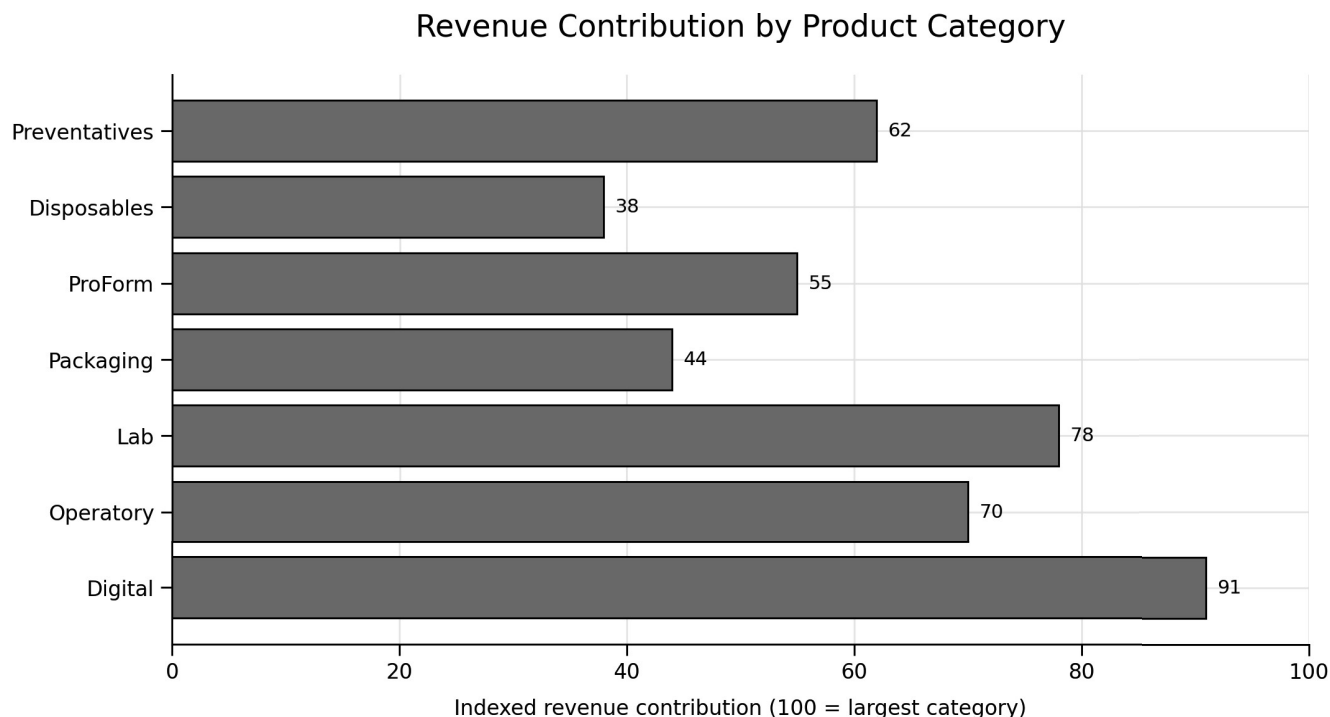
Building a reusable reporting model

1. Start with the ERP-sourced dataset	Use the Power BI download as the controlled source for item, sales, cost, volume, hierarchy, and inventory measures.
2. Standardize the measures	Normalize field names, formats, cost logic, margin calculations, and time periods so reports use consistent definitions.
3. Add business classification	Apply organizational hierarchies and supplemental customer/product logic to distinguish regular inventory from customer-specific or private-label products.
4. Integrate inventory and profitability	Connect on-hand quantity and value with volume, sales, cost, and gross margin at the item level.
5. Build business-unit outputs	Summarize the structured data into focused reports for sales, operations, finance, and leadership.
6. Validate and reuse	Compare average cost with standard cost, review exceptions, and retain the model so future requests can be answered from the same foundation.

A well-designed table can serve many audiences. The same validated structure can support a broad portfolio review, a business-unit report, a customer-specific request, or a private-label analysis.

Revenue contribution by product category

Once the data is structured, category-level reporting makes portfolio concentration visible. The chart below uses indexed synthetic data, where the largest category equals 100. Indexed visuals preserve confidentiality while demonstrating how leadership can compare the relative scale of business segments.



Management interpretation: Relative contribution should guide where leaders spend analytical time, but size alone is not a sufficient performance measure. Large categories may carry lower margins, higher inventory exposure, or greater customer concentration.

Revenue tells management where the business is concentrated. Profitability determines whether that concentration is creating value.

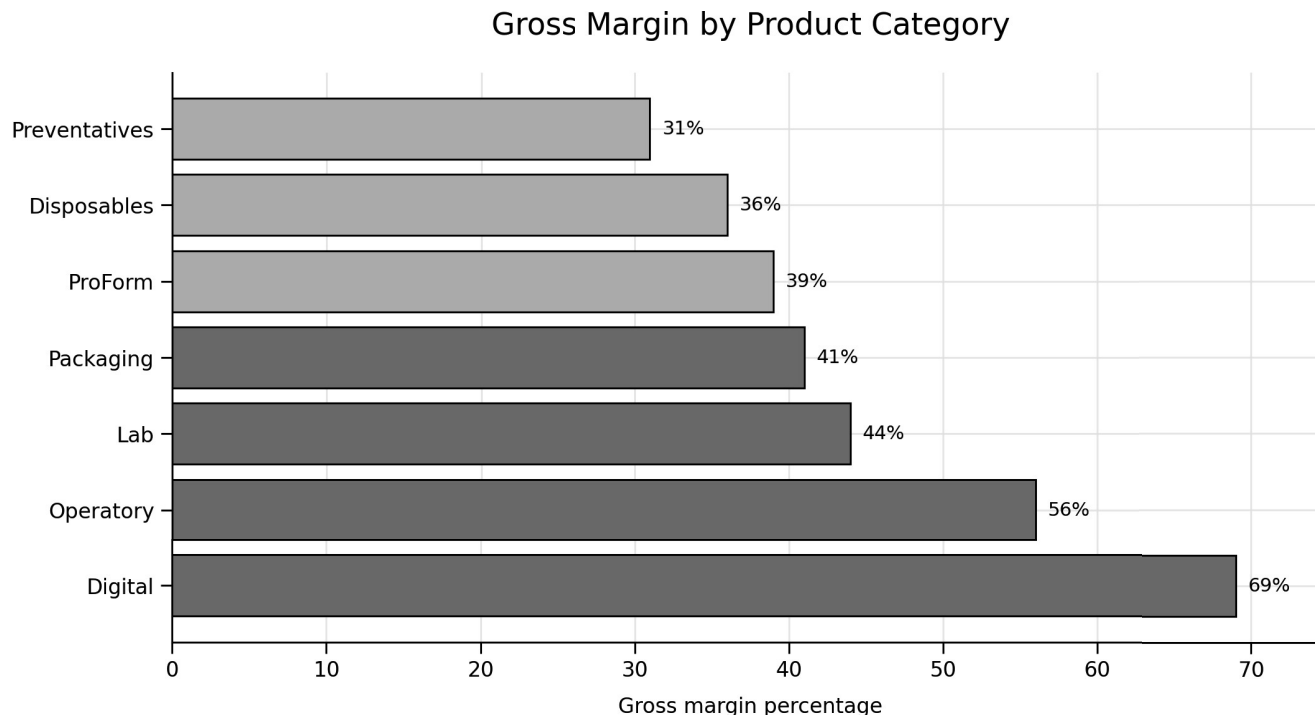
Questions enabled by this view

- Which categories account for the greatest share of revenue?
- Which smaller categories may deserve investment because profitability is stronger?
- Where would a sales decline have the greatest portfolio impact?
- Which categories require deeper customer, SKU, or inventory analysis?

Illustrative example using synthetic data. Categories and indexed values do not represent a specific company or client.

Gross margin by product category

Connecting sales to cost creates a second management lens. In this synthetic example, categories below 40% gross margin are highlighted in gold, making lower-margin areas visually distinct without implying that a single margin threshold is appropriate for every business.



Management interpretation: Margin differences can reflect pricing, customer mix, product mix, material cost, manufacturing complexity, freight, or commercial strategy. The next step is not simply cutting lower-margin business. The next step is understanding the driver and the strategic role of each category.

The combination of revenue scale and margin performance helps separate “big” businesses from businesses that create attractive economic value.

Questions enabled by this view

- Which categories require pricing or cost review?
- Are high-margin categories receiving sufficient commercial attention?
- Do current inventory levels align with demand and profitability?
- Which categories contain a large number of negative- or low-margin SKUs?

Illustrative example using synthetic data. Categories and percentages do not represent a specific company or client.

From ad hoc request to decision-ready report

SITUATION	A sales leader requested customer-specific and private-label information covering current inventory, historical sales, volume, cost, and gross margin.
CONSTRAINT	The information existed across a large ERP-sourced dataset, but the desired management view was not available as a standard report.
ACTION	A reusable item-level analysis was built from the Power BI download. The dataset was summarized, classified, validated, and pivoted into specific business-unit and customer-focused reports.
OUTPUT	The final report presented on-hand inventory, average and standard cost, prior-year performance, current-year performance, and margin in one view.
VALUE	Because the analytical structure already existed, the leadership request could be answered quickly without recreating the analysis from the beginning.

Why this matters

The project demonstrates a repeatable consulting pattern: organize the detailed data once, define business logic explicitly, connect operational and financial measures, and reuse the model to answer new questions. This approach improves consistency and shifts analyst time away from data collection toward interpretation and action.

The report was the visible deliverable. The reusable analytical model was the durable business asset.

What other businesses can apply

1

Begin with the decision

Define the leadership question before selecting fields, visuals, or software.

2

Build one trusted item-level structure

A consistent analytical grain prevents conflicting totals and supports multiple reporting views.

3

Connect operational and financial measures

Inventory becomes more actionable when demand, sales, cost, and margin are visible beside it.

4

Treat classification as business logic

Customer-specific, private-label, business-unit, and product-family rules should be explicit and repeatable.

5

Design for exceptions

Negative margins, low-margin SKUs, unusual cost differences, and inventory without demand should be easy to isolate.

6

Reuse the foundation

A structured model should support recurring dashboards and unplanned leadership questions alike.

Better analytics does not require more reports. It requires a clearer path from raw data to business action.

Practical analytics for stronger business decisions

ML Business Analytics helps organizations transform operational and financial data into clear, actionable information. The focus is on practical solutions that leadership teams can understand, use, and maintain.

Inventory Analytics	Visibility into on-hand value, aging, demand, working capital, and exceptions.
Profitability Analysis	Customer, product, category, pricing, cost, and gross-margin insight.
Reporting & Dashboards	Excel and Power BI reporting designed around management decisions.
Process Improvement	Repeatable reporting structures that reduce manual work and improve consistency.

Have a reporting challenge, profitability question, or inventory problem that your current systems do not answer clearly? Start with the business decision, then build the analytics around it.

START A CONVERSATION

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Methodology and narrative are based on anonymized professional experience. Charts use synthetic sample data. This publication does not disclose client, employer, customer, product-level, pricing, cost, or confidential financial information.